

PAPER Hyperion as a Strategic-Materials Platform: Gain, Tritium Breeding, He-3 Co-Production, and 14 MeV Neutron Output from a Compact Spherical-Tokamak

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Abstract

Fusion breeders have historically foundered on two coupled obstacles: net electricity forces a large plasma and a large blanket, and a self-sufficient closed fuel cycle at power-plant scale implies a dangerous standing tritium inventory. Both are consequences of demanding electricity from a device whose neutrons are physically its most valuable output. We present the integrated physics of Hyperion, a compact negative-triangularity spherical-tokamak breeder sized to a strategic-materials mission — tritium, helium-3, and 14 MeV neutrons — and show that once the net-electricity requirement is dropped, both obstacles fall below their binding thresholds. The frozen design point ($I_p = 9.66$ MA, on-axis $B_0 = 8.0$ T, centrepost peak field 16.84 T, $\delta = -0.30$, $\kappa = 2.0$, $R_0 = 1.20$ m, $A = 2.5$) closes its zero-dimensional power balance at a plasma gain $Q = 3.076$ and fusion power $P_{\text{fus}} = 85.04$ MW, carried openly as a driven, sub-ignition design-point target reached at demonstrated confinement. We state the governing equations of every subsystem as explicit mathematics — the Bosch–Hale power balance, the free-boundary Grad–Shafranov equilibrium and its ideal-MHD margins, the Sauter neoclassical bootstrap, the nonlinear-gyrokinetic transport ratio, the neutron-transport tritium-breeding functional, the tritium and helium-3 mass-balance ordinary differential equations, and the Rosenbluth–Putvinski runaway avalanche — and evaluate them on a verification-first computational stack (Kronos toolkit, FreeGS/FreeGSNKE, NEO, nonlinear CGYRO, TGYRO, ASCOT5, OpenMC on ENDF/B-VIII.0, and a NIMROD-verified disruption model), with the nonlinear gyrokinetics and Monte-Carlo neutronics executed on the NVIDIA GPU HPC campaign. Negative triangularity is the confinement lever: converged nonlinear gyrokinetics show it suppressing ion-scale transport by 40–45% and electron-scale (ETG) transport by 79–80% relative to a positive-triangularity twin, with the operating point subcritical. The free-boundary equilibrium is Mercier-, ballooning- and external-kink-stable with a $1.64\times$ normalized-beta margin to the no-wall Troyon limit; the current is 85% externally driven with a 15.5% bootstrap fraction; fusion-born alphas are 99.98% confined; and tritium breeding rises along a confirmed ladder from a net surplus of 0.29 to ~ 2.0 kg-T yr⁻¹ per unit at the advanced blanket (net breeding ratio 1.42), making a single advanced unit a tritium exporter and a national-scale 4 kg-T yr⁻¹ supply an explicit fleet property. The inboard centrepost is a quantified consumable cartridge, and the disruption response is answered directly and is demanding: a severe runaway-electron avalanche makes shattered-pellet mitigation a hard design requirement. Every headline number is a frozen anchor with a gate serial, independently re-run and stamped in the accompanying physics de-risking register.

Keywords: spherical tokamak; tritium breeder; helium-3; negative triangularity; materials platform; 14 MeV neutron source; consumable centrepost; disruptions; bootstrap current; Grad–Shafranov equilibrium

Nomenclature

Evidence tags: [D] derived, [T] declared, [R] retrieved, [Q] open. Every headline value carries its gate serial (e.g. BR-L1-A1); the serials index the physics de-risking register^[R0], deposited at doi:10.5281/zenodo.22133057.

1 Introduction

Fusion breeders are not a new idea, and they have failed for a consistent reason. To make net electricity a device must be large — a large plasma to hold the power, a large blanket to intercept

Symbol	Definition
Q	plasma gain, $P_{\text{fus}}/P_{\text{ext}}$
I_p, B_0	plasma current; on-axis toroidal field
A, δ, κ	aspect ratio; triangularity (negative here); elongation
$\langle \sigma v \rangle_{DT}$	Maxwellian-averaged D–T reactivity (Bosch–Hale)
β_N	normalized beta, $\beta [\%] aB_0/I_p$
Δ^*	Grad–Shafranov elliptic operator
$F(\psi)$	poloidal current function, RB_ϕ
f_{bs}	bootstrap current fraction
L_{31}, L_{32}, L_{34}	Sauter neoclassical transport coefficients
Q_s, Q_{gB}	species heat flux; gyro-Bohm normalising flux
TBR	net tritium breeding ratio on the 14.06 MeV spectrum
Λ_T	per-unit tritium burn rate (kg-T fpy^{-1})
S	net tritium surplus per unit, $(\text{TBR} - 1) \Lambda_T$
λ_T	tritium decay constant, $\ln 2/t_{1/2}$ ($t_{1/2} = 12.32 \text{ yr}$)
t_{sh}	tungsten-carbide inboard-shield thickness
τ_{CQ}	current-quench time
E_c	Connor–Hastie critical (runaway) electric field
f_α	alpha fraction of fusion energy, $E_\alpha/E_{DT} = 0.200$
fpy	full-power year ($3.156 \times 10^7 \text{ s}$)

it — and to close its own fuel cycle at that scale it must hold a standing tritium inventory measured in kilograms, with all the licensing, safety, and proliferation weight that inventory carries [33, 24]. Both burdens are downstream of one requirement: net electricity from a device whose neutrons are, physically, the most valuable thing it makes. Strip that requirement away, size the machine instead to the *materials* its neutrons produce — tritium for the fusion economy, helium-3 for national-security and quantum applications, and a fast 14 MeV neutron flux for isotope production and materials qualification — and the two classical obstacles both drop below the thresholds that stopped the earlier concepts. The plasma can be compact, and the per-unit tritium inventory a fraction of a power plant’s.

Hyperion is the compact spherical tokamak (ST) that results. A spherical tokamak confines a large plasma current in a small volume behind a high-field inboard centrepost, which is exactly what makes it an efficient neutron source and breeder platform [1, 2, 3]. We choose negative triangularity ($\delta = -0.30$) because it accesses good core confinement without the edge pressure pedestal and the edge-localised-mode cycle of conventional H-mode [4, 5, 6]; the quiet, ELM-free edge is an operational asset for a nuclear device whose first wall cannot tolerate repeated transient loads. The result is a machine that closes its power balance at demonstrated confinement, with the plasma gain carried honestly as a design-point target rather than a demonstrated one.

Contributions. This paper (i) states the governing physics of every subsystem as explicit mathematics and fixes a self-consistent compact operating point at $Q = 3.076$, $P_{\text{fus}} = 85.04 \text{ MW}$ (§2, §4); (ii) describes the verification-first computational stack and the role of the NVIDIA GPU HPC campaign (§3); (iii) demonstrates, with converged nonlinear gyrokinetics, that negative triangularity suppresses turbulence at both the ion and the electron scale into a quiet subcritical core (§5); (iv) shows the point is an ideal-MHD-stable free-boundary equilibrium sustained at an 85%-driven current (§6–§7); (v) confines the fusion-born alphas (§8); (vi) develops the tritium breeding ladder and the mass balance that makes a single advanced unit a net exporter (§9); (vii) treats the centrepost as a quantified consumable and answers the disruption question directly (§10–§11); and (viii) places helium-3 and the strategic products, and shows the deuterium–helium-3 loop does not close indigenously (§12). Every number is a frozen anchor, independently re-run and stamped in the de-risking register^[R0].

Scope. This is a strategic-materials paper. Its success metric is fusion gain, breeding ratio, helium-3, and 14 MeV neutrons — not megawatts to a grid. No net-electricity claim is made for the breeder, and no economic analysis appears here; economics is reported separately and is not part of the public record of this work.

2 Governing equations and the formal problem

The design is the simultaneous solution of a coupled set of physics constraints. We state each governing equation here so that every headline number in §4–§12 traces to a defined quantity rather than an assertion.

2.1 Fusion power balance and gain

On a deuterium–tritium plasma of total density $n = n_D + n_T$ with $n_D = n_T = n/2$, the volumetric fusion power is

$$P_{\text{fus}} = \int_V n_D n_T \langle \sigma v \rangle_{DT}(T_i) E_{DT} dV = \frac{1}{4} \overline{n^2 \langle \sigma v \rangle_{DT}} E_{DT} V, \quad E_{DT} = 17.59 \text{ MeV}, \quad (1)$$

where $\langle \sigma v \rangle_{DT}(T_i)$ is the Maxwellian-averaged reactivity and the overbar denotes a profile-weighted volume average. We evaluate $\langle \sigma v \rangle_{DT}$ with the Bosch–Hale parameterization [9], valid over 0.2–100 keV,

$$\langle \sigma v \rangle_{DT} = C_1 \theta \sqrt{\frac{\xi}{m_r c^2 T_i^3}} \exp(-3\xi), \quad \theta = \frac{T_i}{1 - \frac{T_i(C_2 + T_i(C_4 + T_i C_6))}{1 + T_i(C_3 + T_i(C_5 + T_i C_7))}}, \quad \xi = \left(\frac{B_G^2}{4\theta} \right)^{1/3}, \quad (2)$$

with Gamow constant $B_G = 34.3827 \text{ keV}^{1/2}$, reduced mass $m_r c^2 = 1.124656 \times 10^6 \text{ keV}$, and the tabulated $C_1 \dots C_7$; the curve peaks near 64 keV (Fig. 2). The fraction of fusion energy carried by the alpha particle,

$$f_\alpha = \frac{E_\alpha}{E_{DT}} = \frac{3.52}{17.59} = 0.200, \quad (3)$$

heats the plasma; the complementary 0.800 leaves as 14.06 MeV neutrons — the device’s product. Writing the plasma stored energy $W = \frac{3}{2} \int_V n(T_e + T_i) dV$ and the energy-confinement time $\tau_E = W/P_{\text{loss}}$, the steady-state power balance is

$$\underbrace{f_\alpha P_{\text{fus}}}_{\text{alpha heating}} + P_{\text{ext}} = P_{\text{loss}} + P_{\text{rad}} = \frac{W}{\tau_E} + P_{\text{rad}}, \quad (4)$$

and the plasma gain is $Q \equiv P_{\text{fus}}/P_{\text{ext}}$. Equations (1)–(4) close the zero-dimensional operating point once the confinement quality (the H -factor multiplying a confinement scaling for τ_E) and the profiles are fixed; the resulting frozen values are given in §4.

2.2 Free-boundary equilibrium and stability limits

The axisymmetric equilibrium is the Grad–Shafranov equation for the poloidal flux $\psi(R, Z)$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi), \quad F = R B_\phi, \quad (5)$$

solved as a free-boundary problem so that the coil currents, not a prescribed boundary, set the shape. The relevant global stability limit is the Troyon normalized-beta limit [19]: with $\beta = 2\mu_0 \langle p \rangle / B_0^2$ and

$$\beta_N = \frac{\beta [\%] a B_0}{I_p [\text{MA}]}, \quad (6)$$

ideal-MHD sets a no-wall ceiling β_N^{max} ; the operating margin is $\beta_N^{\text{max}}/\beta_N$. Local interchange and ballooning stability are tested by the Mercier criterion $D_M > 0$ and the s – α ballooning condition on each flux surface, and the low- n external kink by an energy-principle metric $\delta W > 0$.

2.3 Neoclassical bootstrap current

A solenoid-free ST must sustain its current non-inductively, and the pressure-gradient-driven bootstrap current sets the floor. We use the Sauter neoclassical formulation [16], in which the flux-surface-averaged parallel bootstrap current is

$$\langle j_{\parallel} B \rangle_{\text{bs}} = -I(\psi) p_e \left[L_{31} \frac{p}{p_e} \frac{\partial \ln p}{\partial \psi} + L_{32} \frac{\partial \ln T_e}{\partial \psi} + L_{34} \alpha \frac{p_i}{p_e} \frac{\partial \ln T_i}{\partial \psi} \right], \quad (7)$$

with the coefficients $L_{31}, L_{32}, L_{34}(\nu_\star, \epsilon)$ and $\alpha(\nu_\star)$ functions of collisionality and inverse aspect ratio; the bootstrap fraction is $f_{\text{bs}} = I_{\text{bs}}/I_p$, and the balance $I_p - I_{\text{bs}}$ is supplied by external current drive.

2.4 Turbulent transport and the triangularity lever

Core confinement is set by drift-wave turbulence, governed by the nonlinear gyrokinetic equation for the perturbed distribution δf_s of species s ,

$$\frac{\partial \delta f_s}{\partial t} + (v_{\parallel} \mathbf{b} + \mathbf{v}_d + \mathbf{v}_E) \cdot \nabla \delta f_s + v_{\parallel} \frac{\partial \delta f_s}{\partial v_{\parallel}} = C[\delta f_s] + S_s, \quad \mathbf{v}_E = \frac{c}{B} \mathbf{b} \times \nabla \langle \phi \rangle, \quad (8)$$

closed by the gyrokinetic Poisson–Ampère field equations, where $\mathbf{v}_d$ is the magnetic drift and C the collision operator. The turbulent heat flux of species s , $Q_s = \langle \int d^3v \frac{1}{2} m_s v^2 \delta f_s \mathbf{v}_E \cdot \nabla \psi \rangle$, is reported in gyro-Bohm units $Q_{\text{gB}} = n_e T_e c_s (\rho_s/a)^2$ (with $c_s = \sqrt{T_e/m_i}$, $\rho_s = c_s/\Omega_i$). The design lever is the ratio of the saturated flux between a negative- and a positive-triangularity plasma at matched gradients,

$$\mathcal{R}_s = \frac{Q_s^{\delta < 0}}{Q_s^{\delta > 0}} \Big|_{\text{matched } a/L_T}, \quad s \in \{i, e\}, \quad (9)$$

evaluated separately at the ion (ITG+TEM) and electron (ETG) scales.

2.5 Neutronics and the tritium breeding functional

The neutron field obeys the steady linear Boltzmann transport equation

$$\boldsymbol{\Omega} \cdot \nabla \phi(\mathbf{r}, E, \boldsymbol{\Omega}) + \Sigma_t(\mathbf{r}, E) \phi = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}, E' \rightarrow E, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) \phi(\mathbf{r}, E', \boldsymbol{\Omega}') dE' d\boldsymbol{\Omega}' + \frac{S_n(\mathbf{r})}{4\pi}, \quad (10)$$

with a 14.06 MeV source S_n set by the neutron production of (1). The net tritium breeding ratio is the tritium produced per source neutron,

$$\text{TBR} = \frac{1}{S_n} \int_V \int_0^\infty [\Sigma_{\text{Li}}^{(n, \alpha)t}(\mathbf{r}, E) + \Sigma_{\text{Li}}^{(n, n' \alpha)t}(\mathbf{r}, E)] \Phi(\mathbf{r}, E) dE dV, \quad (11)$$

with $\Phi = \int_{4\pi} \phi d\boldsymbol{\Omega}$ the scalar flux; the ${}^7\text{Li}$ channel and beryllium $(n, 2n)$ multiplication set whether $\text{TBR} > 1$.

2.6 Tritium and helium-3 mass balance

The per-unit tritium burn rate follows from (1): one triton is consumed per E_{DT} of fusion, so

$$\Lambda_T = \frac{P_{\text{fus}}}{E_{DT}} m_T (3.156 \times 10^7 \text{ s}) = 4.76 \text{ kg-T fpy}^{-1} \text{ per unit}, \quad (12)$$

and the in-vessel tritium inventory evolves as

$$\frac{dN_T}{dt} = (\text{TBR} - 1) \dot{N}_{\text{burn}} - \lambda_T N_T, \quad \lambda_T = \frac{\ln 2}{t_{1/2}}, \quad t_{1/2} = 12.32 \text{ yr}, \quad (13)$$

whose net surplus rate is $S = (\text{TBR} - 1) \Lambda_T$. Helium-3 is bred not in the blanket but through the beta decay of banked surplus tritium,

$${}^3\text{H} \xrightarrow{\beta^-, t_{1/2}=12.32 \text{ yr}} {}^3\text{He}, \quad \frac{dN_{{}^3\text{He}}}{dt} = \lambda_T N_T, \quad (14)$$

so the helium-3 production rate is capped by the decay of the banked inventory — a slow source, as §12 makes quantitative.

2.7 Disruption dynamics and the runaway avalanche

A disruption terminates the current on the current-quench time τ_{CQ} ; the induced toroidal electric field can seed and then avalanche a runaway-electron beam. The threshold is the Connor–Hastie critical field

$$E_c = \frac{n_e e^3 \ln \Lambda}{4\pi \varepsilon_0^2 m_e c^2}, \quad (15)$$

and above it the runaway population grows by close-collision avalanche [27, 28],

$$\frac{dI_{\text{RE}}}{dt} \sim \frac{I_{\text{RE}}}{\tau_a}, \quad \tau_a^{-1} \propto \frac{E/E_c - 1}{\ln \Lambda}, \quad (16)$$

so that the number of e -foldings scales with $(E/E_c - 1)$ times the current-decay time. As §11 shows, at $E/E_c \approx 1128$ this avalanche is severe, and the mitigation requirement is a first-class design fact.

3 Configuration, codes, and computational methodology

3.1 Configuration

Hyperion is a compact ST: major radius $R_0 = 1.20$ m, minor radius $a = 0.48$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$, on-axis field $B_0 = 8.0$ T with a centrepost peak field of 16.84 T, plasma current $I_p = 9.66$ MA, and negative triangularity $\delta = -0.30$ (Fig. 1). The configuration label is 22021; it is the frozen design point of the series. The high inboard field is delivered by a no-insulation REBCO centrepost magnet, treated in the companion conductor paper [44]; it is a distinct device from, and operates at a lower field than, the 26.49 T plug coil of the separate D-³He tandem-mirror burner products [45]. The two magnets are never conflated.

3.2 Verification-first computational stack

Every headline quantity is a frozen anchor carried in a declared-parameter registry, regenerated by a named script, and tagged by evidence class. The physics codes and their gate serials are: the Kronos toolkit reduced-order evaluator for the zero-dimensional power and particle balance (gate BR-L1-A1); the free-boundary Grad-Shafranov solvers FreeGS and FreeGSNKE [17, 18] for the equilibrium (BR-L1-A12); Mercier, s - α ballooning and a reduced external-kink metric for ideal-MHD stability, with the full $n=1$ energy-principle solve carried as owed work (BR-L2-A19); the drift-kinetic code NEO [15] for the neoclassical bootstrap current (BR-L2-A6); nonlinear CGYRO [11, 10] at ion and electron scale for turbulent transport (BR-L2-A1c, REG-P13), with TGYRO [12] for the flux-driven profile check (BR-L1-A11); the ASCOT5 guiding-centre orbit follower [20, 21] for alpha confinement (BR-L2-A20); OpenMC [22] on ENDF/B-VIII.0 [23] for neutronics, breeding, damage, and activation (BR-L1-A13, BR-L1-A13b, BR-L2-A7u); and a reduced-order disruption model on a NIMROD-verified [25] equilibrium, cross-referenced to DREAM [26] for the runaway population (BR-L2-A10).

3.3 The NVIDIA GPU HPC campaign

The computationally dominant workloads — the converged nonlinear gyrokinetic transport runs (ion-scale at the real electron-mass ratio and electron-scale ETG), the Monte-Carlo neutron transport for the breeding, damage, and centrepost-life scans, and the orbit-following alpha study — were executed on the NVIDIA GPU HPC campaign, an A100/H100/A10 accelerator fleet. Nonlinear CGYRO is a spectral Eulerian solver whose flux-tube and global runs map efficiently onto GPU accelerators; the neutronics scans are embarrassingly parallel over independent particle histories and shielding-thickness cases. The reduced-order evaluators, the free-boundary equilibrium, and the neoclassical solve run on commodity CPU. A subset of the physics gates — and, for the neutronics, the breeding ratio — was independently re-run on separate hardware and on an independent nuclear-data evaluation (ENDF/B-VIII.0 versus FENDL), reproducing the frozen values within stated tolerance. The full per-gate provenance, including the byte-level record and the honest gates with their named next experiments, is the physics de-risking register^[R0] [36].

4 The design point and power balance

Solving (1)–(4) at the frozen configuration returns $Q = 3.076$ at $P_{\text{fus}} = 85.04$ MW (Table 1, gate BR-L1-A1). Of the fusion power, $f_\alpha = 0.200$ ((3)) is retained as alpha heating and 0.800 leaves as 14.06 MeV neutrons (Fig. 3); the neutron power is 68.0 MW and the volumetric source strength $S_n = 3.01 \times 10^{19} \text{ ns}^{-1}$ (gate BR-OUT-3). The neutron wall loading is 1.885 MW m^{-2} and the first-wall damage rate $18.85 \text{ dpa fpy}^{-1}$ (BR-L1-A13).

We carry the gain honestly as a *design-point target*: no device has exceeded $Q \approx 0.7$ (Fig. 4), and the value is reached at demonstrated confinement rather than demonstrated on a machine. Critically, the target is a driven, sub-ignition gain — a materially easier requirement than net electricity, and the one a breeder actually needs, since alpha heating supplies only $f_\alpha Q = 0.62$ of the external power at this point and the plasma is nowhere near an ignition runaway.

5 Confinement: the negative-triangularity lever

The power balance closes only if turbulent transport does not cap the pressure below the operating gradient. Negative triangularity is the design’s confinement lever, and we establish it at both turbulence scales with nonlinear gyrokinetics, computing the flux ratio $\mathcal{R}_s$ of Eq. (9).

5.1 Method

We evolve the nonlinear gyrokinetic equations (8) with CGYRO [11] at the frozen equilibrium, carrying kinetic electrons, and compare a negative-triangularity case against a

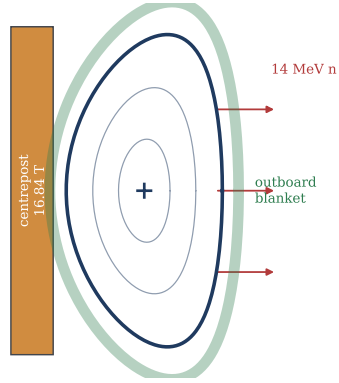
Hyperion poloidal section: $R_0 = 1.20$ m, $a = 0.48$ m, $\kappa = 2.0$, $\delta = -0.30$ 

Figure 1. (Schematic.) Hyperion poloidal cross-section: negative-triangularity ($\delta = -0.30$) separatrix with the D-shape opening outboard, the high-field consumable centrepost (peak 16.84 T) on the inboard side, the outboard breeding blanket, and the 14.06 MeV neutron product. Geometry is to the frozen design point; no field data are plotted.

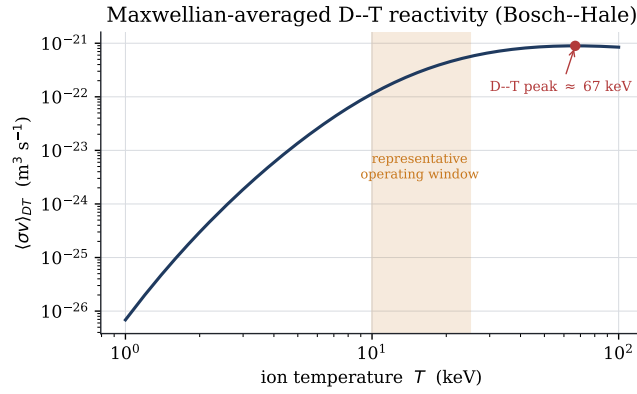


Figure 2. Maxwellian-averaged D–T reactivity $\langle\sigma v\rangle_{DT}(T_i)$ from the Bosch–Hale parameterization [9] of Eq. (2), peaking near 64 keV; the shaded band marks a representative operating window. The reactivity curve is textbook physics; the operating point of Table 1 is the frozen anchor.

Table 1. Hyperion design point (config 22021). Every value is a frozen anchor with its gate serial.

quantity	value	gate
major / minor radius	1.20 / 0.48 m	22021
aspect ratio A / elongation κ	2.5 / 2.0 (2.47 free-boundary)	BR-L1-A12
triangularity δ	–0.30	BR-L1-A7
plasma current I_p	9.66 MA	BR-L1-A1
on-axis field B_0 / centrepost peak	8.0 / 16.84 T	BR-L1-A12
plasma gain Q	3.076	BR-L1-A1
fusion power P_{fus}	85.04 MW	BR-L1-A1
neutron fraction / power	0.800 / 68.0 MW	Eq. (1), BR-OUT-3
source strength S_n	3.01×10^{19} n s ^{–1}	BR-OUT-3
wall loading / damage rate	1.885 MW m ^{–2} / 18.85 dpa fpy ^{–1}	BR-L1-A13
normalized beta β_N	2.13	BR-L1-A12
bootstrap fraction f_{bs}	0.155	BR-L2-A6

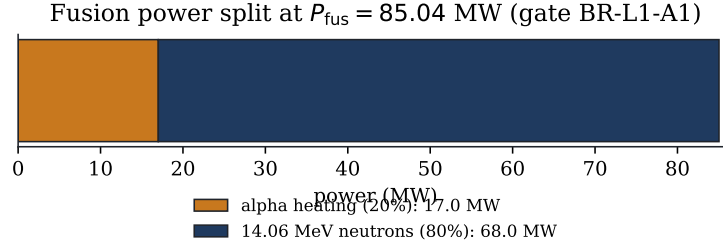


Figure 3. Fusion power split at the design point (Eq. (1), (3)): 20% of the fusion energy is alpha heating retained in the plasma, 80% leaves as 14.06 MeV neutrons — the strategic product.

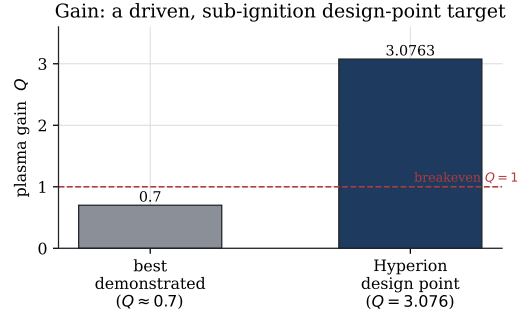


Figure 4. The plasma gain is a design-point target at demonstrated confinement: no device has exceeded $Q \approx 0.7$; Hyperion's driven, sub-ignition target is $Q = 3.076$ (gate BR-L1-A1).

positive-triangularity twin at matched gradients. Ion-scale runs resolve the ITG and TEM branches at the real electron-mass ratio ($\mu = m_i/m_e = 3672$; gate BR-L2-A1c, converged and saturated); electron-scale runs resolve the ETG branch at the standard reduced mass ratio $\mu = 400$ (REG-P13). Transport is reported as the saturated gyro-Bohm-normalized heat flux, and the negative-to-positive ratio $\mathcal{R}_s$ is the headline.

5.2 Results

At the operating gradient the converged ion-scale run returns $Q_i^{\delta>0} = 64.2$ and $Q_i^{\delta<0} = 37.2$ (gyro-Bohm units), a suppression of $\mathcal{R}_i = 0.58$, i.e. 42%, squarely within the frozen 40–45% band (gate BR-L2-A1c); the operating point sits below the nonlinear threshold, so the core is quiet and subcritical, the confinement-favourable regime. At the electron scale, negative triangularity suppresses ETG transport by 79–80% (gate REG-P13): Fig. 5 shows the electron heat flux falling by roughly a factor of five at both sampled gradients. The direction and approximate magnitude are corroborated against the TCV and DIII-D negative-triangularity campaigns [7, 4, 8] and are the subject of a dedicated gyrokinetic companion paper [37], with a two-code quasilinear TGLF cross-check reported separately. A flux-driven TGYRO evaluation returns $\tau_E = 0.41$ s ($H_{98} \approx 0.94$), consistent with the zero-dimensional balance (BR-L1-A11).

6 Free-boundary equilibrium and ideal-MHD stability

A free-boundary Grad–Shafranov solution of Eq. (5) [17, 18] reproduces the design point on a 129×129 grid, with poloidal beta $\beta_p = 0.371$, edge safety factor $q_{95} = 2.90$, internal inductance $\ell_i = 1.40$, and normalized beta $\beta_N = 2.13$ (Eq. (6)). Against the no-wall Troyon limit $\beta_N^{\max} \approx 3.5$ this is a factor 1.64 of margin (gate BR-L1-A12). The equilibrium is Mercier- ($D_M > 0$) and ballooning-stable on all confinement surfaces ($q > 1$; a core $q < 1$ sawtooth pocket is the only exception), and the low- n external kink carries a margin of 1.15 – $1.31\times$ on the reduced metric evaluated over $\beta_N = 2.13$ – 2.44 against a no-wall limit of ≈ 2.8 (BR-L2-A19; Fig. 6). Two items are flagged rather than smoothed: the free-boundary elongation reaches $\kappa = 2.47$, above the nominal 2.0, coupling to vertical stability; and the kink margin is a reduced metric pending a full resistive-wall $n=1$ energy-principle (δW) solve. The equilibrium and stability study is a companion paper [38].

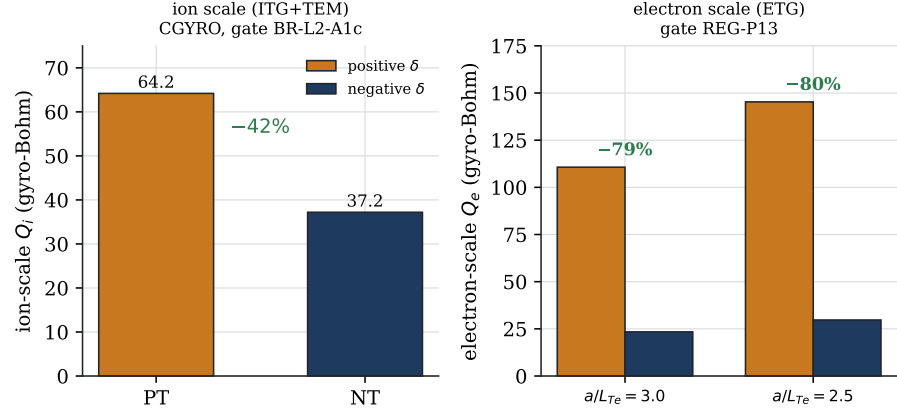


Figure 5. Negative-triangularity (NT) versus positive-triangularity (PT) turbulent transport. **Left:** converged ion-scale (ITG+TEM) heat flux from nonlinear CGYRO at the real mass ratio (gate BR-L2-A1c) — a 42% suppression. **Right:** electron-scale (ETG) saturated heat flux at two normalized gradients (gate REG-P13) — a 79–80% suppression. The operating point is subcritical at both scales.

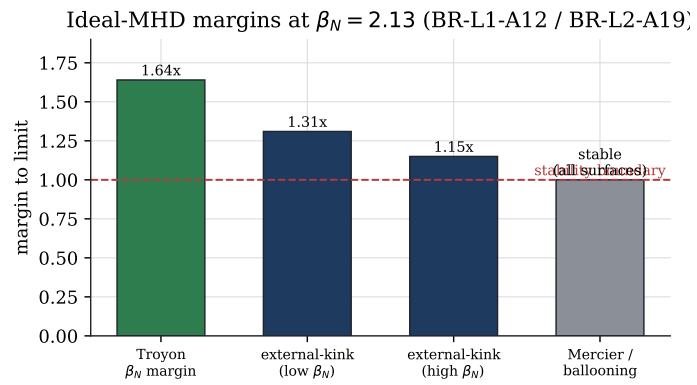


Figure 6. Ideal-MHD margins at $\beta_N = 2.13$ (gates BR-L1-A12, BR-L2-A19): the Troyon normalized-beta margin (1.64 $\times$), the external-kink margin over the $\beta_N = 2.13$ –2.44 band (1.31–1.15 $\times$), and Mercier/ballooning stability on all confinement surfaces. All quantities clear the stability boundary.

7 Current drive, bootstrap, and sustainment

A solenoid-free ST must sustain its current non-inductively. Evaluating the Sauter bootstrap integral (7) with NEO [15] at the design profiles returns $f_{bs} = 0.155$ (1.50 MA; gate BR-L2-A6), so the plasma current $I_p = 9.66$ MA is 85% externally driven (8.16 MA) plus the 15.5% bootstrap (Fig. 7). The driven fraction is supplied by neutral-beam and radio-frequency current drive; the bootstrap fraction, while modest at this beta, aligns favourably with the negative-triangularity pressure profile. The steady-state current-drive study — source technology, coupling, and the path to a higher bootstrap fraction — is developed separately.

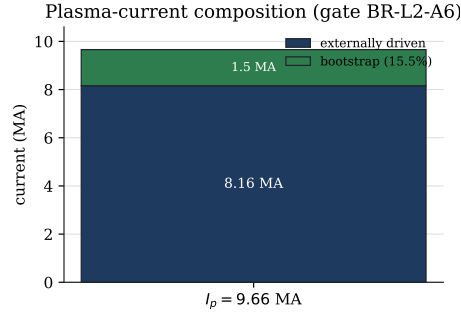


Figure 7. Plasma-current composition (gate BR-L2-A6): 8.16 MA externally driven plus a 1.50 MA (15.5%) bootstrap fraction from the Sauter neoclassical evaluation of Eq. (7).

8 Fast-ion and alpha-particle confinement

Fusion-born alphas must deposit their energy in the plasma and must not concentrate on the first wall. A guiding-centre orbit-following calculation with ASCOT5 [20, 21] over 10^5 markers returns 99.98% confinement with a prompt loss of 0.023% and a mean first-wall alpha load of 0.083 kW m^{-2} (peak 1.94), far below any engineering limit (gate BR-L2-A20). Finite toroidal-field ripple opens a second channel but keeps the loss sub-percent (Fig. 8): 0.019% at 1.0% ripple and 0.031% at 1.5%, with the peak wall load of order 1 kW m^{-2} . This closes a prior analytic estimate that had suggested a $\sim 4.5\%$ loss; the orbit-following result refutes it and agrees with an independent GPU evaluation (BR-L1-A14). Alpha confinement is not a limiting concern; the detailed study is a companion paper [39].

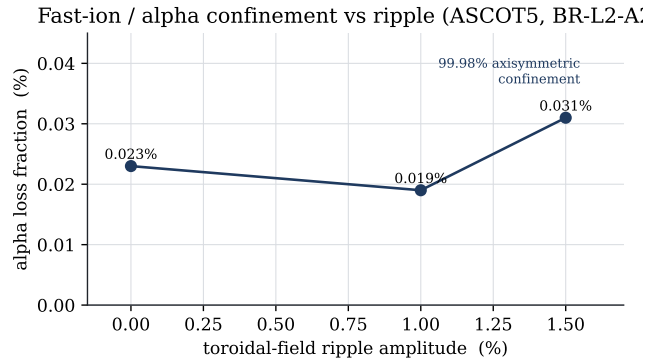


Figure 8. Alpha loss fraction versus toroidal-field ripple amplitude (gate BR-L2-A20, ASCOT5). Even at 1.5% ripple the loss stays sub-percent against 99.98% axisymmetric confinement.

9 Blanket, tritium, and the breeding ladder

Tritium breeding is a design lever along a confirmed ladder, not a fixed number. The per-unit burn rate is $\Lambda_T = 4.76 \text{ kg-T fpy}^{-1}$ from Eq. (12) (gate BR-OUT-2), and the net surplus per unit is $S = (\text{TBR} - 1) \Lambda_T$ from the mass balance (13). Evaluating the breeding functional (11) with OpenMC on ENDF/B-VIII.0 across blanket configurations (gate BR-L2-A7u) traces the ladder of Fig. 9: the literal frozen recipe gives net TBR ≈ 1.06 and $S = 0.29$; a dedicated 5 cm solid-beryllium multiplier raises it to TBR ≈ 1.19 and $S = 0.9$; and the advanced blanket — a void-reduced breeder with a solid-beryllium multiplier and a steel reflector — reaches net TBR = 1.42 and

$S \approx 2.0 \text{ kg-T yr}^{-1}$ per unit. This TBR = 1.42 is the confirmed per-unit ceiling across twelve blanket configurations: a lithium–lead alternative does not beat it, and the ST centrepost caps the inboard coverage that sets the ceiling. The cross-library check (ENDF/B-VIII.0 versus FENDL) reproduces the breeding ratio within tolerance and is the load-bearing verification of the export claim.

A single advanced unit is therefore a tritium exporter at the $\sim 2.0 \text{ kg-T yr}^{-1}$ scale. A national-scale 4 kg-T yr^{-1} supply is explicitly a *fleet* number ($3.6\text{--}4.5 \text{ kg-T yr}^{-1}$ from four to five units, or two advanced units), never a single-unit claim. The blanket and fuel-cycle study, and the fleet path to self-sufficiency, are companion papers [40].

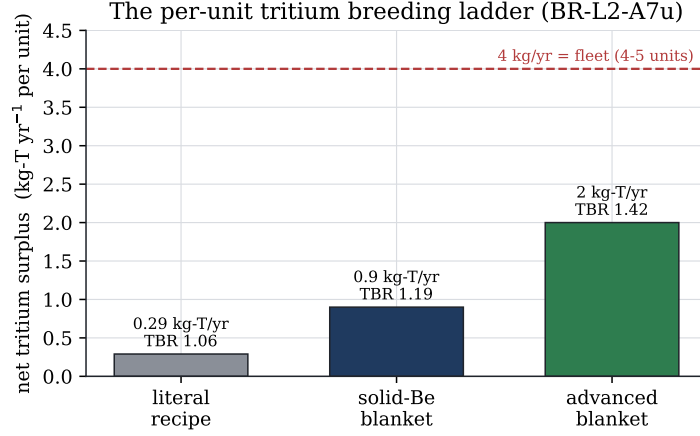


Figure 9. The per-unit tritium breeding ladder (gate BR-L2-A7u, Eqs. (11)–(12)): net surplus $S = (\text{TBR} - 1)\Lambda_T$ at the literal, solid-beryllium, and advanced blankets. The advanced-blanket TBR = 1.42 is the confirmed per-unit ceiling; 4 kg-T yr^{-1} is a fleet number.

10 The consumable centrepost

The inboard centrepost sits closest to the source and is the machine’s dominant lifetime risk. Rather than pretend it is a lifetime component, the design treats it as a consumable, scheduled-replacement cartridge. Fig. 10 shows the coverage-versus-protection trade from OpenMC (gate BR-L1-A13b): as the tungsten-carbide inboard shield thickens from 5 to 40 cm, net TBR falls from 1.196 to 1.097 while the centrepost life at the REBCO fast-fluence threshold ($3.9 \times 10^{22} \text{ n m}^{-2}$) rises from 0.0048 to 0.194 fpy; at the most-shielded design point (a full 45 cm shield, cutting the fast fluence from 2.9×10^{25} to $1.7 \times 10^{23} \text{ n m}^{-2} \text{ fpy}^{-1}$) the best-case cartridge life is 0.58 fpy. The trade is steep and binding: no split simultaneously delivers a high TBR and a multi-year centrepost, which is precisely why the consumable frame is correct. The coil integrity, the fused quench detector, and the maintenance cadence are quantified in a companion paper [41].

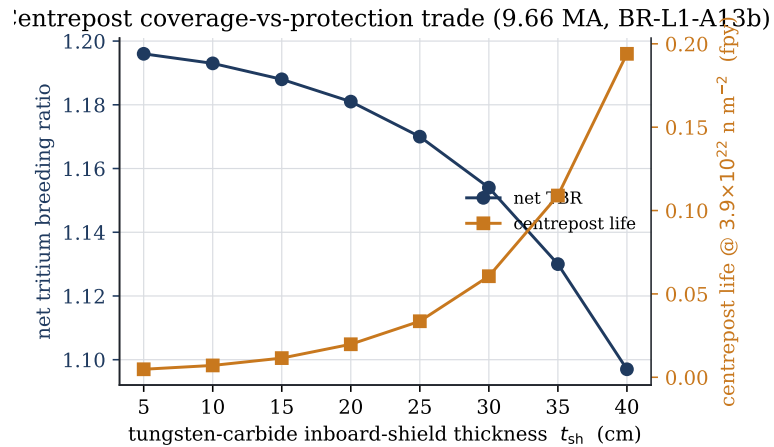


Figure 10. Centrepost coverage-versus-protection trade at 9.66 MA (gate BR-L1-A13b): net TBR (left axis) and centrepost life at the $3.9 \times 10^{22} \text{ n m}^{-2}$ fast-fluence threshold (right axis) versus tungsten-carbide inboard-shield thickness. The trade is monotone and binding.

11 Disruptions, runaway electrons, and electromagnetic loads

The disruption question is answered directly, and its answer is demanding. On a NIMROD-verified equilibrium [25] the current-quench time is $\tau_{CQ} = 7\text{--}19\text{ ms}$ (gate BR-L2-A10), which sets the mitigation-trigger speed. The runaway-electron avalanche is severe: with the toroidal field far above the Connor–Hastie critical field of Eq. (15), $E/E_c \approx 1128$, the avalanche gain of Eq. (16) amplifies a milliampere seed to a multi-megaampere beam carrying of order 70% of the plasma current [27, 28, 26]. The halo current is 35% of I_p , setting the vessel and first-wall mechanical load basis. The load-bearing conclusion is carried as a first-class design fact: shattered-pellet (or equivalent) disruption mitigation is a hard requirement, with a trigger inside τ_{CQ} ; the risk is not claimed negligible. The disruption and mitigation study, including the mitigation-effectiveness case, is a companion paper [42], and the community mitigation basis is the ITER programme [29, 30].

12 Helium-3 and the strategic products

Helium-3 is bred independently of the tritium blanket — through the decay of surplus tritium at equilibrium, Eq. (14) — and is the machine’s strategic co-product at about $0.9\text{ kg}\cdot\text{Tyr}^{-1}$ per unit (gate BR-OUT-2), comparable to the entire current US domestic supply. It is not a bulk fuel source for the companion burner: the 12.32-year tritium half-life makes decay-sourced helium-3 far too slow relative to a burner’s consumption. Solving the fuel-loop balance (gate BR-OUT-4) against a burner net demand of $170.2\text{ kg}\cdot^3\text{He fpy}^{-1}$ gives a closure ratio of $\sim 192/105/45$ breeders per burner at the three blanket rungs (Fig. 11); the deuterium–helium-3 loop does not close indigenously, and burner-scale helium-3 is lunar-gated around 2038–40 [32, 43]. The strategic value of the breeder’s helium-3 — in national-security, cryogenic, and quantum applications — is real and distinct from bulk fuel supply, and beyond helium-3 the same 14 MeV flux co-produces medical and industrial isotopes. We keep strategic value separate from revenue.

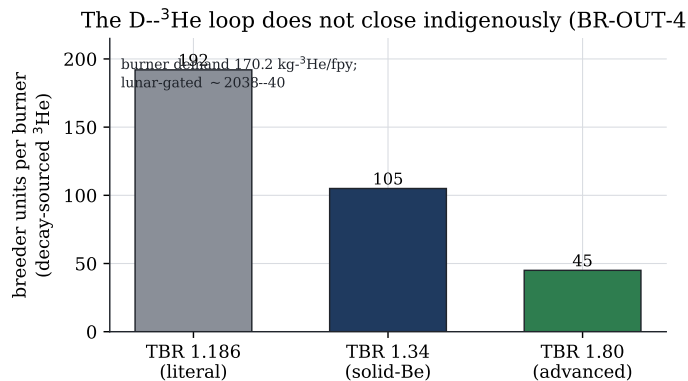


Figure 11. The D– ^3He loop does not close indigenously (gate BR-OUT-4): breeder units required per burner from decay-sourced helium-3 (Eq. (14)) at the three blanket rungs, against a burner demand of $170.2\text{ kg}\cdot^3\text{He fpy}^{-1}$. Burner-scale helium-3 is lunar-gated $\sim 2038\text{--}40$.

13 Relation to prior fusion-breeder concepts

Fusion-breeder concepts are not new; what is new is a design that places both classical obstacles below threshold at once (Table 2). Prior tritium-assisted and deuterium–helium-3 breeder proposals [33, 34, 35] were sized to net-electric or satellite-power missions and inherited large machine scale or large tritium inventory. By sizing to a strategic-materials mission and adopting negative triangularity for confinement and a consumable centrepost for lifetime, Hyperion reaches a compact operating point at a driven sub-ignition gain, with a per-unit tritium inventory inside civil licensing precedent [24]. The comparison is qualitative by necessity — the prior concepts differ in fuel, confinement, and mission — but the pattern is clear: the obstacles that stopped the earlier concepts are the ones the mission reframe removes.

14 Discussion

Sized to a materials mission rather than to electricity, the compact negative-triangularity breeder balances its power at demonstrated confinement, with both historical breeder obstacles — scale and tritium inventory — below their binding thresholds. It is worth reading each subsystem against the published benchmarks. The confinement lever is quantified at both turbulence scales

Table 2. Hyperion in the context of prior fusion-breeder concepts (qualitative).

concept	mission	binding obstacle	Hyperion's answer
D–T assisted breeder [33]	net power	machine scale	compact ST, mission reframe
IEC D– ³ He breeder [34]	satellite ³ He	low power density	magnetic ST, $Q = 3.076$
tandem-mirror breeder [35]	satellite power	plug potential	(separate burner papers)
Hyperion	strategic materials	— (below threshold)	compact, consumable centrepost

with converged nonlinear gyrokinetics, and the direction and magnitude of the negative-triangularity suppression are consistent with the TCV and DIII-D experimental campaigns [7, 4, 8, 6]; the ion-scale result sits at the confirmed end of the 40–45% band rather than the transiently larger figures that appear before saturation. The equilibrium is ideal-MHD-stable with a $1.64\times$ Troyon margin (Eq. (6)), within the operational-limit envelope established for tokamaks [19, 30]; the bootstrap fraction from the Sauter formulation (7) is modest, as expected at this beta, and the sustainment burden falls on external drive. The alpha-confinement result (99.98%) is comfortable against any first-wall limit and refutes a prior analytic over-estimate. The breeding ladder places the advanced-blanket ceiling at $TBR = 1.42$; against the tritium-fuel-cycle physics of a power plant [24], the key move is not a higher single-unit ratio but the mission reframe that makes a modest surplus a strategic export while keeping the per-unit inventory small. The disruption assessment is the most demanding result and is carried as such: the runaway avalanche of Eq. (16) at $E/E_c \approx 1128$ makes mitigation a hard requirement, consistent with the runaway and mitigation literature [27, 28, 29]. Finally, the helium-3 mass balance is honest about what decay-sourced helium-3 can and cannot do: it is a strategic co-product, not a burner fuel supply, and the loop is lunar-gated [32].

15 Limitations

The gain is a design-point target; no device has demonstrated it. The ion-scale gyrokinetics are converged at the real mass ratio, but the all-in-one multi-scale run is a nice-to-have rather than a closed gate. The kink margin is a reduced metric pending a full energy-principle calculation, and the free-boundary elongation–vertical-stability coupling at $\kappa = 2.47$ is flagged. The bootstrap fraction is modest and the driven-current sources are stated rather than engineered here. The advanced-blanket ceiling requires the advanced blanket to be built, and the disruption assessment is reduced-order on a verified equilibrium. Each item is stated with its evidence class and has a named next step in the register^[R0].

16 Conclusion

Hyperion is a compact negative-triangularity spherical-tokamak breeder that reaches $Q = 3.076$ at 85.04 MW on a mission of strategic materials. Its governing physics is stated as explicit mathematics and evaluated on a verification-first stack, with the nonlinear gyrokinetics and Monte-Carlo neutronics on the NVIDIA GPU HPC campaign. Negative triangularity suppresses turbulence at both scales into a quiet subcritical core; the equilibrium is ideal-MHD-stable and non-inductively sustained; alphas are 99.98% confined; the centrepost is a consumable cartridge; disruptions demand mitigation; and a single advanced unit exports tritium at the $\sim 2.0 \text{ kg}\cdot\text{Tyr}^{-1}$ scale while co-producing helium-3 and a fast neutron flux. Freed from a net-electricity requirement, the machine places both classical breeder obstacles below threshold, and the hard part — the power balance — is solved at demonstrated confinement. It closes on the frozen physics model as a platform; the distance from closed-on-model to closed-demonstrated is a named experiment — the measured breeding ratio at first-of-a-kind operation ($\sim 2030\text{--}31$) — not open physics.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the reviewers of the series and Dr Patrick H Diamond for critical review, and the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. Every headline quantity is a frozen anchor (BR-L1-A1, BR-L1-A12, BR-L2-A19, BR-L2-A1c, BR-L2-A20, BR-L2-A6, BR-L2-A7u, BR-L1-A13b, BR-L2-A10, BR-OUT-2/3/4) in the KRONOS Physics De-Risking Register^[R0], doi:10.5281/zenodo.22133057, and is regenerable from the open, CC-BY-4.0 deposit; this paper is archived at doi:10.5281/zenodo.22645691; official version:

<https://www.kronosfusionenergy.com/publications/breeder.html>. No result depends on unpublished or proprietary data.

Data availability

The figure data (power split, reactivity curve, turbulence-suppression ratios, current composition, alpha-loss scan, breeding ladder, centrepost trade, stability margins, and fuel-loop closure) are regenerable from the deposited figure script and trace to the register^[R0] by the gate serials cited inline. This paper is archived at doi:10.5281/zenodo.22645691, which references the Kronos 2026 series. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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A Design parameter set

Table 3 collects the frozen design parameters with their gate serials; the same values drive the figures and the companion papers.

Table 3. Frozen design parameters (config 22021).

group	parameter	value
geometry	$R_0 / a / A / \kappa / \delta$	1.20 m / 0.48 m / 2.5 / 2.0 / -0.30
plasma	$I_p / B_0 / \text{centrepost} / Q / P_{\text{fus}}$	9.66 MA / 8.0 T / 16.84 T / 3.076 / 85.04 MW
stability	$\beta_N / \text{Troyon margin} / \text{kink}$	2.13 / $1.64 \times$ / $1.15\text{--}1.31 \times$
equilibrium	$\beta_p / q_{95} / \ell_i$	0.371 / 2.90 / 1.40
current	driven / bootstrap (f_{bs})	8.16 MA / 1.50 MA(0.155)
confinement	NT ion / electron suppression	40–45 % / 79–80 %
alpha	confinement / wall load	99.98 % / 0.083 kW m^{-2}
breeding	burn / advanced TBR / surplus	$4.76 \text{ kg-T fpy}^{-1}$ / 1.42 / $\sim 2.0 \text{ kg-T yr}^{-1}$
centrepost	best-case life / damage	0.58 fpy / $18.85 \text{ dpa fpy}^{-1}$
disruption	$\tau_{\text{CQ}} / \text{halo} / E/E_c$	7–19 ms / 35 % I_p / 1128
helium-3	per unit / fuel-loop closure	0.9 kg-T yr^{-1} / $\sim 192/105/45$ breeders per burner